

Retail Sales Performance Analysis

Executive Summary

This project analyzes retail sales data to identify revenue trends, customer purchasing behavior, product performance, and regional sales distribution. The dataset was cleaned and preprocessed using Python (Pandas), analyzed using PostgreSQL and SQL, and visualized through an interactive Power BI dashboard. The insights can help management improve marketing strategies, inventory planning, and customer retention.

Business Problem

The retail business aims to improve sales performance and customer understanding by answering the following business questions:

- How do sales change over time?
- Which regions generate the highest and lowest sales?
- Which product categories contribute the most revenue?
- Which sales representatives perform best?
- How profitable is each region?
- What data-driven insights can support better business decision-making?

Objectives

The main objectives of this project are:

- Clean and prepare raw retail sales data for analysis.
- Handle missing values and standardize the dataset.
- Analyze the cleaned dataset using SQL to answer key business questions.
- Analyze sales trends, regional performance, product performance, and sales representative performance.
- Build an interactive Power BI dashboard to visualize KPIs and business metrics.
- Generate business insights and recommendations to support data-driven decisions.

Dataset Overview

- Dataset Name: Retail Sales Dataset
- Records: 1,000
- Features: 15 (reduced columns after cleaning)
- Source: CSV file
- Description: Contains customer, product, payment, store, quantity, price and sales information.

Tools Used

- Python (Pandas)
- PostgreSQL
- SQL
- Power BI

Project Methodology / Steps Involved

Imported the retail sales dataset (CSV) into Python for data cleaning and preprocessing.

Data Cleaning & Preprocessing:

- Inspected the dataset (`head`, `info`, `describe`)
- Checked missing values.
- Checked duplicates.
- Removed unnecessary columns.
- Standardized column names.
- Converted data types.
- Converted Date to datetime.
- Feature engineering (`year`, `month`, `age_group`)
- Loaded cleaned dataset into PostgreSQL

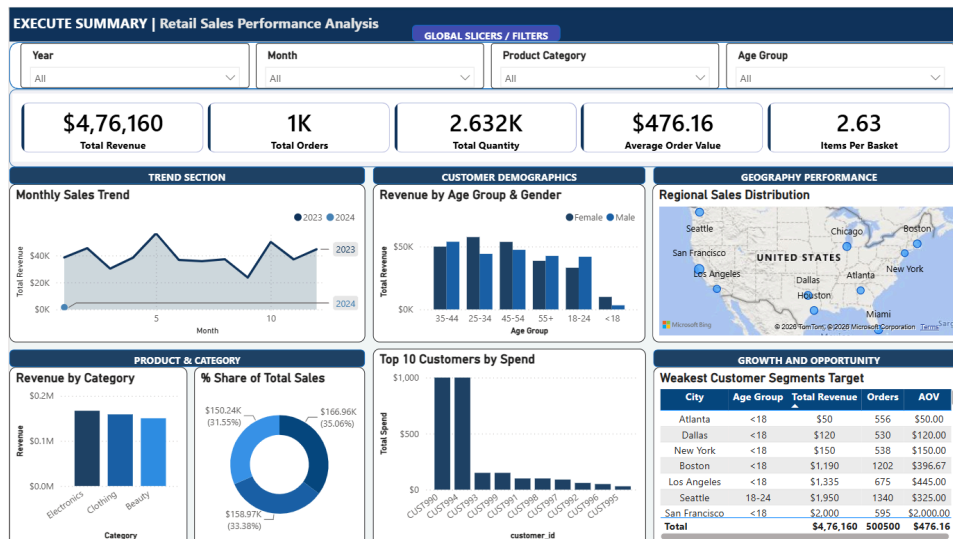
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Business questions were answered using SQL to analyze:

- Monthly Sales Trends
- City-wise Performance
- Category-wise Revenue
- Revenue and Average Order Value
- Customer Demographics
- Top Customers
- Weak Customer Segments

Dashboard Creation

Developed an interactive Power BI dashboard featuring KPI cards, sales trend analysis, regional performance, product category analysis, sales representative performance, profit analysis, and interactive slicers for dynamic reporting.



Key Findings

- **Sales Trends:** Sales show seasonal fluctuations.
- **Regional Performance:** San Francisco generated the highest revenue, while New York generated the lowest.

- **Product Performance:** Electronics generated the highest revenue.
- **Customer Insights:** Female customers aged 25–34 contributed the highest revenue.
- **Business Opportunity:** Regional and demographic analysis identified opportunities for targeted marketing.

Recommendations

- Increase inventory before high-sales months.
- Improve marketing in lower-performing cities.
- Launch loyalty programs for top customers.
- Target promotions toward customers aged 25–44.
- Continue investing in Electronics while promoting Beauty products to increase average order value.

Conclusion

This project demonstrates practical skills in data cleaning, SQL-based analysis, and interactive dashboard development, showcasing the complete end-to-end workflow of a Data Analyst. By transforming raw retail sales data into meaningful business insights, the project highlights the ability to support data-driven decision-making and solve real-world business problems.